PrefBench: Benchmarking Interactive Personal Agents via Preference Fit and Cross-Preference Generalization

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Abstract

Interactive personal agents are increasingly trained and evaluated by interacting with LLM-based user simulators. However, existing methods are often compared under heterogeneous simulators, task setups, and evaluation criteria, making it difficult to determine whether a method truly learns user decision preferences or merely exploits artifacts of a particular simulation environment. Recent work further shows that LLM user simulators can diverge from real humans both behaviorally and evaluatively, raising concerns about the reliability of simulator-based conclusions.

We propose PrefBench, a unified benchmark and evaluation framework for *interactive preference learning* in personal agents. Our key idea is to reinterpret simulator-based interaction not as open-ended dialogue quality assessment, but as learning a latent distribution over user decision preferences. Under this view, a method should be evaluated along two principal axes: (1) *preference fit*, i.e., whether the agent can recover a user's decision preference through multi-turn interaction, and (2) *cross-preference generalization*, i.e., whether the learned interaction strategy transfers to unseen users and unseen preference families. PrefBenchoperationalizes this perspective through preference-centric task construction, train/test splits defined over users or profile-induced preference families, metrics based on decision agreement and interaction efficiency, and a human calibration protocol that estimates the simulator-to-human transferability of benchmark conclusions.

Rather than claiming to solve the full simulator-to-human gap in user simulation, PrefBenchtargets a narrower but central problem: providing a more unified, objective, and task-grounded way to evaluate interactive personal agents that learn user decision preferences through interaction.

1 Introduction

Personal agents are increasingly expected to make or support decisions on behalf of users in domains such as recommendation, planning, scheduling, shopping, and information triage. In these settings, high-quality behavior depends not only on world knowledge or tool use, but also on the ability to *learn a particular user's decision preferences through interaction*. For example, a travel assistant may need to infer whether a user prioritizes cost over convenience, whether they are risk-averse to delays, or whether they prefer brand familiarity over novelty. These latent preferences are rarely fully specified upfront and often must be elicited adaptively over multiple turns.

A common experimental paradigm is to evaluate such agents by letting them interact with LLM-based user simulators. This paradigm is attractive because it is scalable, inexpensive, and supports closed-loop evaluation. However, the resulting literature remains difficult to compare for two reasons. First, different works use different user simulators, prompts, task constructions, and metrics, making cross-paper comparison unreliable. Second, recent evidence suggests that LLM user simulators may not be faithful proxies for real users: they can be overly cooperative, reveal

information too early, and produce evaluative judgments that deviate from human perceptions [2, 3, 8]. As a result, agent improvements in simulation may not correspond to improvements with real users.

For the specific problem of *interactive preference learning*, we argue that current evaluation practice is misaligned with the task's core objective. What fundamentally matters is not whether the dialogue "sounds good" in a generic sense, nor whether a single rule-based reward is triggered at the end of interaction. Instead, the central questions are: Can the agent *recover the user's decision preference*? How efficiently can it do so? And does its learning strategy generalize to users or preference types not seen during development?

In this paper, we introduce PrefBench, a unified benchmark for interactive preference learning agents. Our starting point is a conceptual reframing: simulator-based interaction should be viewed as fitting a latent distribution over user decision preferences. Under this lens, we evaluate agents using two main capabilities:

1. **Preference fit**: whether the final decision made by the assistant aligns with the user's true preference under the given context;
2. **Cross-preference generalization**: whether the agent's interactive strategy transfers to unseen users and unseen preference families.

To operationalize this view, PrefBenchintroduces: (i) preference-centric task construction, where each task instantiates a user, a context, and a decision space; (ii) train/test splits defined over users or profile-induced preference families rather than only over tasks; (iii) metrics based on decision agreement, ranking similarity, regret, and interaction efficiency; and (iv) a human calibration stage that measures whether simulator-based method rankings transfer to real users.

Our work is motivated by, but distinct from, recent critiques of simulator-to-human mismatch. We do *not* claim to solve the full behavioral realism problem of LLM user simulation. Instead, we focus on a narrower but practically important objective: providing a more unified, objective, and task-grounded benchmark for interactive personal agents that learn decision preferences through interaction. In this sense, PrefBenchis best understood as a *Sim2Real-aware evaluation framework* for preference-learning agents, not a complete solution to user simulation fidelity.

Our main contributions are:

1. We propose a unified benchmark formulation for *interactive preference learning*, centered on preference fit and cross-preference generalization rather than subjective dialogue quality alone.
2. We introduce evaluation protocols that split train and test by *users* and by *preference families*, enabling systematic measurement of adaptation to unseen users and unseen preference structures.
3. We define a metric suite covering final decision alignment, preference recovery quality, interaction efficiency, and simulator-to-human transfer.
4. We present a human calibration protocol for validating whether simulator-based conclusions are predictive of real-user outcomes.

2 Related Work

User simulation for interactive systems. User simulation has a long history in dialogue systems, where rule-based and statistical simulators were used for policy learning and evaluation [7]. With the rise of large language models, recent work has increasingly adopted LLMs as user simulators for evaluating conversational systems and agents [1, 4, 6]. This line of work has substantially improved

the scale and flexibility of interactive evaluation, but has also introduced a new dependency on the fidelity of simulators themselves.

Reliability of LLM-based user simulators. A growing body of work questions whether LLM simulators are faithful proxies for human behavior or human judgment [2, 3, 5]. Most directly relevant to our motivation, Zhou et al. [8] show that LLM user simulators can diverge from real humans both as interactive users and as evaluators, often creating an "easy mode" that inflates apparent agent performance. Our work is aligned with these concerns, but differs in scope: rather than aiming to fully close the simulator-to-human gap, we focus on building a more objective benchmark for a specific class of tasks—interactive learning of user decision preferences.

Interactive preference learning and personal agents. A broad range of systems seek to infer user preferences from interaction, including personalized recommenders, planning assistants, and decision-support agents. Yet evaluation remains fragmented. Some work reports downstream task success; some uses synthetic preference labels; some uses ad hoc simulator prompts; others rely on subjective human rating. We argue that this fragmentation obscures the core question of interest: whether an agent can *recover a user's latent decision preference efficiently and robustly*. PrefBenchaims to provide a common evaluation layer for such methods.

Benchmark design for generalization. Benchmarking by random task splits can overestimate generalization when train and test share users, styles, or latent patterns. In personalized systems, a key distinction is whether a method generalizes to *new tasks for the same user*, *new users from known preference families*, or *new preference families altogether*. Our split design makes these distinctions explicit and measurable.

3 Problem Setup

We formalize interactive preference learning as sequential decision-making under latent user preferences.

3.1 Users, contexts, and decisions

Let $u \in \mathcal{U}$ denote a user, $x \in \mathcal{X}$ a decision context, and $\mathcal{Y}(x)$ a set of candidate decisions or actions available under context x. The user is associated with a latent preference function

$$f_u(x, y) : \mathcal{X} \times \mathcal{Y}(x) \to \mathbb{R}, \tag{1}$$

which scores how desirable decision y is for user u in context x.

The user's observable behavior may take several forms:

- **Single-choice preference**: selecting one option from $\mathcal{Y}(x)$;
- **Ranking preference**: producing an ordering over options;
- **Pairwise preference**: expressing which of two options is preferred;
- **Attribute trade-off**: revealing weights over features such as cost, convenience, brand, novelty, or risk.

In all cases, the agent does not observe f_u directly. Instead, it interacts with the user to infer enough of the latent preference structure to make a high-quality final decision.

3.2 Interactive preference elicitation

At each turn $t \in \{1, \ldots, T\}$, the assistant chooses an interaction action a_t (e.g., a clarifying question, a comparison query, or a proposed option), and the user returns a response r_t. The interaction history up to turn t is

$$h_t = (x, a_1, r_1, \ldots, a_t, r_t). \tag{2}$$

After at most T turns, the assistant outputs either:

- a final decision $\hat{y} \in \mathcal{Y}(x)$, or
- an estimated preference model $\hat{f}_u$, from which a final decision is derived.

The core evaluation problem is therefore not generic dialogue quality, but the quality and efficiency of recovering the user's decision preference through interaction.

3.3 Preference families

To study generalization, we assume that each user belongs to a preference family $z(u) \in \mathcal{Z}$, where a family captures a broader preference pattern such as price sensitivity, risk aversion, novelty seeking, or convenience prioritization. Preference families can be defined in several ways:

- directly from structured user profiles;
- induced from historical decision behavior via clustering;
- defined by annotation rules over preference attributes.

This abstraction allows us to distinguish generalization to unseen *users* from generalization to unseen *types of preferences*.

4 PrefBench: A Unified Benchmark for Interactive Preference Learning

We now present PrefBench, our proposed benchmark and evaluation framework.

4.1 Design principles

PrefBenchis guided by four principles.

P1: Preference-centric evaluation. The primary object of evaluation should be the agent's ability to recover and act on user preference, not only its surface conversational fluency.

P2: Generalization-aware splits. Benchmark splits should separate adaptation to new tasks for known users from adaptation to new users and unseen preference families.

P3: Efficiency as a first-class metric. In personal-agent settings, interaction cost matters. A method that reaches correct decisions with fewer turns or less information burden should be rewarded.

P4: Human calibration. Simulator-based conclusions should be checked against real users on a controlled subset of tasks to estimate transfer reliability.

4.2 Task construction

Each task instance in PrefBenchcontains:

1. a user u with profile information p_u and optional historical traces D_u;
2. a context x specifying the current decision scenario;
3. a candidate decision space $\mathcal{Y}(x)$;
4. a hidden ground-truth preference target, expressed as a label such as a final choice, a ranking, or pairwise preferences;
5. an interactive interface through which the assistant may ask questions and receive user responses.

The benchmark is domain-agnostic. Example instantiations include:

- travel option selection;
- personalized product recommendation;
- schedule planning under trade-offs;
- restaurant or content selection;
- information triage with user-specific priorities.

A key requirement is that the task must support *incremental preference revelation*. If the correct decision can be inferred directly from the initial prompt without ambiguity, the task does not meaningfully evaluate interactive preference learning.

4.3 User simulators as preference environments

We use LLM-based user simulators to instantiate scalable interaction environments. However, unlike prior paradigms that often use the simulator both to *generate* interaction and to *judge* it holistically, PrefBenchtreats the simulator primarily as a *preference environment*.

Concretely, a simulator receives:

- the user profile p_u;
- optional historical behavior D_u;
- the current context x;
- the hidden preference target or preference-generating logic;
- the interaction history h_t.

It then produces the next user response r_{t+1}.

To reduce degenerate "easy-mode" behavior, the simulator should satisfy two operational constraints:

Consistency constraint. The simulator should maintain a coherent underlying preference structure across turns and across semantically similar contexts.

Gradual revelation constraint. The simulator should not reveal all relevant information immediately unless naturally warranted. Certain information should remain latent until elicited; uncertainty and ambiguity should be allowed.

These constraints do not fully solve behavioral realism, but they help ensure that the benchmark tests preference learning rather than exploitation of over-cooperative simulators.

4.4 Preference diversity and simulator sanity checks

Because PrefBenchrelies on simulated users to instantiate preference distributions, simulator quality must be validated at least at the level of *decision behavior*. We therefore include the following sanity checks:

Inter-user diversity. Under the same context x, users with different profiles should produce meaningfully different choices or rankings.

Intra-user consistency. For semantically related contexts, the same user should exhibit stable preference tendencies unless the context explicitly changes the trade-off structure.

Preference-family coverage. The simulator should cover multiple preference families rather than collapse onto a narrow default behavior.

Response-level non-degeneracy. Responses should not systematically reveal all attributes upfront, nor should they reduce every interaction to trivial one-turn disambiguation.

4.5 Train/test splits

PrefBenchdefines three evaluation settings.

Setting A: Seen-user adaptation. Train and test share users but not task instances. This setting measures whether a method can use prior exposure to a user's preference structure when facing new contexts.

Setting B: Unseen-user, seen-family generalization. Test users are disjoint from train users, but their preference families have appeared during training:

$$\mathcal{U}_{\text{train}} \cap \mathcal{U}_{\text{test}} = \emptyset, \qquad \mathcal{Z}_{\text{test}} \subseteq \mathcal{Z}_{\text{train}}. \tag{3}$$

This measures whether the agent generalizes to new individuals within familiar preference regimes.

Setting C: Unseen-family generalization. Test users belong to preference families absent from training:

$$\mathcal{Z}_{\text{train}} \cap \mathcal{Z}_{\text{test}} = \emptyset. \tag{4}$$

This is the strongest generalization test and aims to evaluate whether the agent has learned a robust interaction strategy rather than memorized user templates.

4.6 Human calibration stage

To estimate whether simulator-based findings transfer to real users, PrefBenchincludes a human evaluation subset. On a carefully sampled set of tasks, real participants interact with the same assistants under the same contexts. We then compare:

- the absolute performance gap between simulation and human evaluation;
- the ranking correlation of methods under simulated and human users;
- the stability of conclusions about interaction efficiency and generalization.

This calibration stage does not replace large-scale simulation, but it provides a principled estimate of transfer reliability.

5 Evaluation Metrics

We now define the metric suite in PrefBench.

5.1 Preference fit

Decision agreement. For single-choice tasks, we compute

$$\mathrm{Acc} = \mathbb{E}\left[\mathbf{1}(\hat{y} = y_u)\right], \tag{5}$$

where y_u is the user's ground-truth preferred decision.

Ranking similarity. For ranking tasks, we report Kendall's τ and Spearman's ρ between the predicted ranking and the user's target ranking.

Pairwise preference accuracy. For pairwise tasks, we compute the fraction of correctly predicted pairwise comparisons.

Utility regret. When a utility representation is available, we compute

$$\mathrm{Regret} = f_u(x, y_u^\star) - f_u(x, \hat{y}), \tag{6}$$

where $y_u^\star = \arg\max_{y \in \mathcal{Y}(x)} f_u(x, y)$ is the user's optimal decision.

5.2 Interaction efficiency

Turns-to-threshold. For a target preference-fit threshold γ, we compute the average number of turns required to reach it:

$$\mathrm{TTT}_\gamma = \mathbb{E}\left[\min\{t : \mathrm{Fit}(h_t) \geq \gamma\}\right]. \tag{7}$$

Token or query cost. We also report the number of user-facing queries, total tokens, or both.

Information efficiency. We define information efficiency as improvement in preference fit per unit interaction cost:

$$\mathrm{InfoEff} = \frac{\Delta\mathrm{Fit}}{\#\mathrm{Turns\ or\ \#Tokens}}. \tag{8}$$

5.3 Generalization

User generalization gap.

$$\Delta_{\text{user}} = \mathrm{Perf}_{\text{seen-user}} - \mathrm{Perf}_{\text{unseen-user}}. \tag{9}$$

Family generalization gap.

$$\Delta_{\text{family}} = \mathrm{Perf}_{\text{seen-family}} - \mathrm{Perf}_{\text{unseen-family}}. \tag{10}$$

Smaller gaps indicate more robust generalization.

5.4 Simulator robustness

Since conclusions may depend on the chosen simulator, we measure performance variance across simulator families:

$$\mathrm{Var}_{\mathrm{sim}} = \mathrm{Var}\left(\mathrm{Perf}^{(1)}, \ldots, \mathrm{Perf}^{(K)}\right), \tag{11}$$

where $\mathrm{Perf}^{(k)}$ is the performance under simulator k.

5.5 Human calibration

Human agreement. We measure the same preference-fit metrics on real users.

Rank transfer correlation. Let π_{sim} and π_{human} denote method rankings in simulation and human evaluation. We report a rank correlation such as Spearman's ρ:

$$\rho_{\mathrm{transfer}} = \rho(\pi_{\mathrm{sim}}, \pi_{\mathrm{human}}). \tag{12}$$

Sim-to-human gap. For any metric M,

$$\Delta_{S2H}(M) = M_{\mathrm{sim}} - M_{\mathrm{human}}. \tag{13}$$

A useful benchmark should not only support high simulator-side discrimination, but also preserve method ordering and qualitative conclusions under human evaluation.

6 Experimental Protocol

This section describes a submission-ready experimental design. In the current draft, numerical results are intentionally omitted and should be filled after running the benchmark.

6.1 Benchmark instantiation

We instantiate PrefBenchon one or more domains where user preferences matter and can be elicited interactively. Candidate domains include:

- shopping or recommendation tasks with multi-attribute trade-offs;
- travel planning tasks balancing cost, time, reliability, and comfort;
- scheduling tasks balancing convenience, constraints, and priorities.

For each task, we require:

1. user-level metadata or profile text;
2. a context with multiple plausible options;
3. a target preference signal (choice, ranking, or pairwise label);
4. sufficient ambiguity to justify interaction.

6.2 Baselines

We recommend comparing at least four classes of methods.

B1: Non-interactive profile-only baseline. The assistant directly predicts a decision from the user profile and task context without asking questions.

B2: Heuristic interactive baseline. The assistant asks a fixed set of templated clarifying questions before making a decision.

B3: Generic LLM interactive agent. A general-purpose LLM determines what to ask and what to decide, without explicit preference-state modeling.

B4: Preference-aware interactive agent. An agent that explicitly maintains a belief or posterior over user preference and selects questions with an information-gain objective.

These baselines disentangle the benefits of interaction from the benefits of active preference modeling.

6.3 User simulators

We recommend using multiple simulator families to test robustness:

- one or more frontier proprietary LLMs;
- one or more strong open-source LLMs;
- optionally, a simulator specialized for user modeling.

Each simulator should be run under the same benchmark protocol, with standardized prompting and hidden preference exposure.

6.4 Human evaluation

We sample a stratified subset of tasks for real-user validation. For each sampled task, real participants are given:

- a role or profile description consistent with the benchmark instance;
- the decision context and available options;
- an interface for interacting with the assistant.

We then collect:

1. the user's final preferred decision;
2. the interaction trace;
3. optional subjective feedback on burden, clarity, and trust.

The purpose of this stage is not to fully replace simulation, but to calibrate whether method comparisons observed in simulation remain predictive under real humans.

6.5 Implementation details

To support reproducibility, all methods should share:

- the same maximum interaction budget T;
- the same option space and context representation;
- the same user profile fields;
- the same stopping criteria;
- the same simulator prompts except for model-specific API wrappers.

We recommend reporting average performance over multiple random seeds, especially when methods involve sampling-based question generation or simulator stochasticity.

Table 1: Main results on PrefBench. Replace `TBD` with actual measurements after experiments.

Method	Seen-U Acc	Unseen-U Acc	Unseen-F Acc	Regret ↓	Turns ↓	Human Corr. ↑
Non-interactive	TBD	TBD	TBD	TBD	–	TBD
Heuristic interactive	TBD	TBD	TBD	TBD	TBD	TBD
Generic LLM agent	TBD	TBD	TBD	TBD	TBD	TBD
Preference-aware agent	TBD	TBD	TBD	TBD	TBD	TBD

Placeholder for a Pareto plot of preference fit vs. interaction cost.
Each point corresponds to one method under one split setting.

Figure 1: Planned visualization of the trade-off between preference fit and interaction cost.

7 Results Template and Planned Analyses

This section provides a draft structure for the empirical results section. It is intentionally written in a paper-ready style, but all quantitative entries should be replaced by actual numbers after running the benchmark.

7.1 Main benchmark results

Table 1 reports the core benchmark results across the three generalization settings.

The main questions to answer are:

1. Does interaction improve preference fit over non-interactive baselines?
2. Do explicitly preference-aware methods outperform generic LLM agents?
3. How large is the degradation from seen users to unseen users and unseen preference families?
4. Which methods preserve their relative ranking under human calibration?

7.2 Preference fit versus interaction cost

A central trade-off in interactive personal agents is whether better fit requires substantially more user burden. Figure 1 should plot preference fit as a function of turns or token cost. Methods that lie on the Pareto frontier are preferable.

7.3 Generalization to unseen users and unseen families

We recommend reporting both absolute performance and generalization gaps:

$$\Delta_{\text{user}}, \quad \Delta_{\text{family}}. \tag{14}$$

Table 2: Simulator robustness. Lower variance indicates that conclusions are less dependent on the chosen user simulator.

Method	Sim-A Acc	Sim-B Acc	Sim-C Acc	Perf Variance ↓
Non-interactive	TBD	TBD	TBD	TBD
Heuristic interactive	TBD	TBD	TBD	TBD
Generic LLM agent	TBD	TBD	TBD	TBD
Preference-aware agent	TBD	TBD	TBD	TBD

A method may achieve strong in-distribution performance yet rely heavily on memorizing user templates, leading to poor transfer to unseen families. This analysis is especially important for methods that condition strongly on profile text.

7.4 Robustness across simulators

Table 2 should quantify how conclusions change across simulator families.

This analysis addresses a practical concern in simulator-based benchmarking: whether a method's apparent superiority is stable or simulator-specific.

7.5 Human calibration

Human calibration should answer the following:

- Do the best methods in simulation remain strong under real users?
- Are efficiency conclusions preserved under human interaction?
- Is there a consistent simulator-to-human gap across methods, or do some methods transfer better than others?

We recommend reporting method ranking correlations and plotting simulated versus human performance for each method.

7.6 Preference diversity and sanity checks

To validate the benchmark itself, we recommend reporting:

1. diversity of decisions across users under fixed contexts;
2. consistency of each simulated user's choices across related contexts;
3. distribution of response lengths and information revelation patterns;
4. ablations on whether removing gradual-revelation constraints makes the benchmark artificially easier.

8 Discussion

Why a preference-centric benchmark? Interactive personal agents are often evaluated using coarse success metrics or subjective judgments that do not directly reflect whether the agent has learned what matters most in personalized decision-making: the user's latent preferences. By shifting evaluation toward preference fit and generalization, PrefBenchmakes the target capability more explicit.

What PrefBenchdoes not claim. PrefBenchdoes not claim that LLM user simulation is fully solved, nor does it eliminate all simulator-to-human mismatch. In particular, user behavior realism remains an open problem. Our claim is narrower: for the class of tasks where the goal is to learn a user's decision preference through interaction, benchmark conclusions should be tied to measurable preference recovery and calibrated against real users wherever feasible.

A bridge between scalable simulation and human validation. A fully human-only benchmark is often too expensive to scale, while a purely simulator-based benchmark may be brittle. PrefBenchis designed as a middle ground: scalable simulation for broad comparison, combined with a targeted human calibration stage for external validity.

9 Limitations

Our proposal has several limitations.

First, user profiles are not identical to preferences. While profile text and historical behavior may provide useful conditioning signals, they do not fully determine a user's latent decision rule. In practice, benchmark builders must be careful not to conflate demographic or descriptive metadata with preference itself.

Second, simulator realism remains a residual source of error. Although PrefBenchintroduces preference-level sanity checks and gradual-revelation constraints, simulated users may still differ from humans in subtle ways.

Third, human calibration is resource-constrained. The human evaluation subset may be too small to estimate absolute deployment performance with high precision, though it can still provide valuable evidence about method ranking stability.

Fourth, benchmark conclusions may depend on domain choice. Preference elicitation in shopping, planning, and recommendation may differ materially from preference elicitation in healthcare, education, or high-stakes advisory tasks.

10 Ethics, Privacy, and Human Subjects

Because PrefBenchconcerns user profiles and personalized decision-making, privacy and human-subject considerations are essential.

Privacy and data minimization. Benchmark instances should avoid exposing unnecessary personal information. When using real user traces, profile fields should be anonymized and minimized to only those needed for the task.

Human evaluation. Any human study should be reviewed under appropriate institutional procedures. Participants should be informed that they are interacting with an AI assistant and that their responses may be used for research evaluation.

Bias and fairness. Preference-family definitions should not encode harmful stereotypes. When grouping users into preference families, one should distinguish between behavioral patterns and sensitive demographic categories.

Personalization risks. A benchmark that rewards aggressive preference inference may encourage systems that over-query users or infer sensitive attributes. For this reason, efficiency and user burden should be treated as core metrics rather than afterthoughts.

11 Conclusion

We introduced PrefBench, a unified benchmark and evaluation framework for interactive preference learning in personal agents. Our central proposal is to evaluate simulator-based interaction not primarily through subjective dialogue quality or single binary rewards, but through the agent's ability to recover user decision preferences, do so efficiently, and generalize across users and preference families. By combining preference-centric tasks, user- and family-level splits, decision-based metrics, simulator robustness analysis, and human calibration, PrefBenchaims to provide a more objective and comparable foundation for evaluating interactive personal agents.

More broadly, we view this benchmark as a step toward a more task-grounded methodology for user-centric AI evaluation: one that preserves the scalability of simulation while making explicit what is being learned, how efficiently it is learned, and whether simulator-side conclusions survive contact with real users.

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A Appendix A: Benchmark Instantiation Checklist

This appendix provides a practical checklist for instantiating PrefBenchon a concrete dataset.

A.1 A.1 Data requirements

A valid benchmark instantiation should specify:

1. the user representation (profile, history, or both);
2. the task context schema;
3. the decision space;
4. the ground-truth preference label type;
5. the mapping from users to preference families.

A.2 A.2 Simulator prompt design

The simulator prompt should explicitly separate:

- hidden preference information;
- publicly available profile information;
- response constraints (e.g., gradual revelation, consistency);
- turn budget and stopping behavior.

A.3 A.3 Human calibration protocol

For each selected task in the human subset:

1. recruit k participants matching the desired user role distribution;
2. present the task context and option set;
3. allow the participant to interact with each method under a counterbalanced design;
4. record final preference and optional qualitative feedback.

B Appendix B: Example Tables to Fill

Table 3: Example ablation table.

Variant	Acc	Regret ↓	Turns ↓	Human Corr. ↑
Full method	TBD	TBD	TBD	TBD
w/o profile	TBD	TBD	TBD	TBD
w/o interaction	TBD	TBD	TBD	TBD
random questioning	TBD	TBD	TBD	TBD
w/o gradual revelation simulator	TBD	TBD	TBD	TBD